ALFIDO TECH

RIDING THE WAVES OF TECHNOLOGICAL ADVANCEMENT

Internship: ALFIDO Tech Data Science Internship
Project: Task 2 - Superstore Sales Data Analysis
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1. Abstract

This study examines a Superstore retail dataset to track overarching sales performance, generate Key Performance Indicators (KPIs), and visualize revenue generation across varying product categories. Using Python (Pandas, Matplotlib, Seaborn), the raw data was cleaned, column headers were standardized, and temporal data was converted for time-series extraction. The analysis successfully identified core business metrics, including a Total Revenue of $2,261,536.78, yielding an Average Order Value (AOV) of $459.48 across 4,922 unique orders.

2. Introduction

Understanding sales patterns is crucial for retail superstores to optimize their inventory, focus marketing efforts, and drive revenue. This report details the programmatic analysis of a superstore's transaction records. Through systematic data cleaning, feature engineering of temporal metrics, and data visualization via bar plots, we identified the total sales volume and broke down the performance by specific product categories and sub-categories to provide a clear picture of the store's financial health.

3. Problem Statement

Dataset:-https://www.kaggle.com/datasets/bhanupratapbiswas/superstore-sales

Goal:- Clean the raw sales data, extract time-based features, calculate prima ry financial KPIs, and visually analyze the sales performance of different product categories and sub-categories to uncover revenue drivers.

4. Objectives

1.Standardize Data: Clean messy column headers by stripping whitespace and formatting them consistently.

2.Temporal Feature Engineering: Convert string-based order and shipping dates into functional datetime objects and extract Year and YearMonth for seasonal analysis.

3.Calculate Key Performance Indicators: Extract top-level financial metrics including Total Revenue, Total Unique Orders, and Average Order Value (AOV).

4.Build Visualizations: Create layered bar charts to visually map out which product sub-categories drive the highest sales volumes.

5. Data Description

Dataset Name: Superstore Final Dataset
The dataset contains transactional records of a retail superstore, encompassing customer details, geographical regions, and specific product sales data
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Key Columns:

- order_id & row_id: Unique identifiers for the transactions.
- order_date & ship_date: Timestamps for transaction and fulfillment .
- customer_name, segment, country, city, state, region: Customer demographic and geographic data.
- category & sub_category: Product classification (e.g., Furniture, Office Supplies, Technology).
- sales: The total numeric revenue generated per line item.

6. Data Preprocessing

Purpose: Transform raw dataset fields into an analysis-ready format by handling inconsistent text formatting and data types.

Key Processing Steps:

1. Header Standardization: Converted all column headers to lowercase, stripped trailing/leading whitespace, and replaced spaces with underscores (e.g., Order_ID to order_id). This allows for seamless programmatic access.
2. Datetime Conversion: Parsed the order_date and ship_date columns using pd.to_datetime(dayfirst=True) to convert them from raw strings to functional datetime objects.
3. Feature Engineering: Created two new analytical columns—YearMonth (formatted to track monthly periods) and Year—to allow for future time-series and seasonal aggregations.

7. Methodology And Implementation

The project was executed in a Jupyter Notebook environment (Task_2 (3).ipynb) using the following libraries:

- Pandas:For dataingestion, headercleaning,featureengineering, and groupby aggregations.
- Matplotlib & Seaborn: For generating data visualizations.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load the dataset
df = pd.read_csv('superstore_final_dataset (1).csv',encoding="latin1")
df.head(10)
```

Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID
1	CA-2017-152156	08-11-2017	11-11-2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-BO-10001798
2	CA-2017-152156	08-11-2017	11-11-2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-CH-10000454
3	CA-2017-138688	12-06-2017	16-06-2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036.0	West	OFF-LA-10000240
4	US-2016-108966	11-10-2016	18-10-2016	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	FUR-TA-10000577
5	US-2016-108966	11-10-2016	18-10-2016	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	Florida	33311.0	South	OFF-ST-10000760

KPI Calculation Logic:

- Total Revenue was calculated by summing the sales column.
- Total Unique Orders was derived by counting the unique order_id values (nunique()).
- Average Order Value was calculated by dividing Total Revenue by the Total Unique Orders.

```python
# Total Revenue
total_revenue = df['sales'].sum()

# Total Unique Orders
total_orders = df['order_id'].nunique()

# Average Order Value (Total Sales / Total Orders)
aov = total_revenue / total_orders

print(f"Total Revenue: {total_revenue:,.2f}")
print(f"Total Unique Orders: {total_orders}")
print(f"Average Order Value: {aov:,.2f}")
```

```
Total Revenue: 2,261,536.78
Total Unique Orders: 4922
Average Order Value: 459.48
```

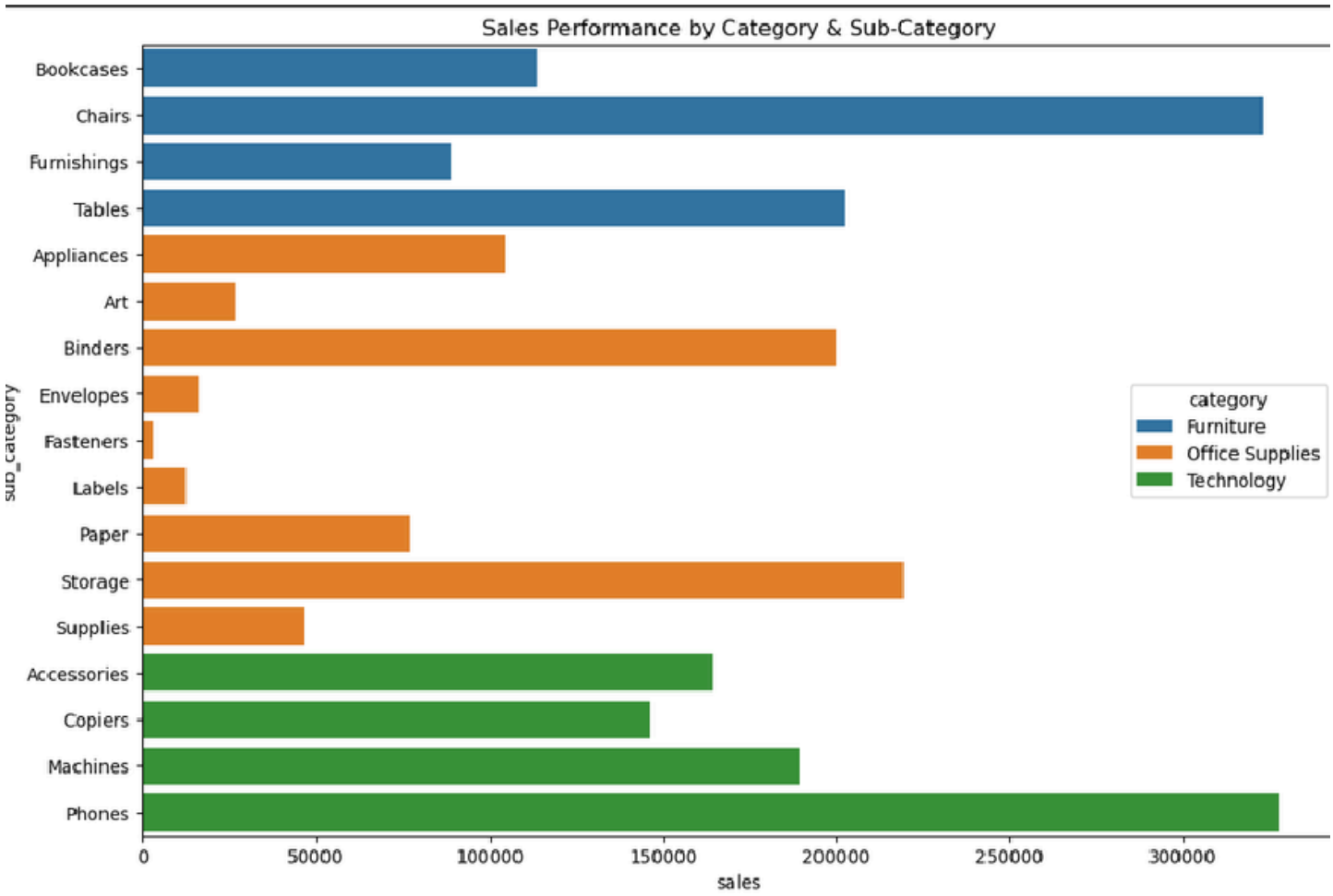
8. Data Visualization

To analyze exactly where the revenue was coming from, the data was grouped by category and sub_category, taking the sum of the sales column.

- SalesbyCategory &Sub-Category: A horizontal barplot (sns.barplot) was generated mapping sales on the x-axis and sub_category on the y-axis, heavily color-coded (hue) by parent category. This visualization allows stakeholders to instantly identify top-grossing products (e.g., Chairs, Phones) versus underperforming items (e.g., Labels, Fasteners).

```
# Sales by Category and Sub-Category
cat_sales = df.groupby(['category', 'sub_category'])['sales'].sum().reset_index()

plt.figure(figsize=(12, 8))
sns.barplot(data=cat_sales, x='sales', y='sub_category', hue='category', dodge=False)
plt.title('Sales Performance by Category & Sub-Category')
```



9. Business Insights

Based on the code execution and output generation, the following key insights were extracted:

1.Macro Financial Health: The superstore processed 4,922 unique orders, generating a massive Total Revenue of $2,261,536.78.

2.High Customer Value: The Average Order Value (AOV) stands at $459.48, indicating that customers are typically making bulk purchases or buying high-ticket items per transaction.

3.Performance Stratification: By visualizing the categorized sales,the business can now identify which specific sub-categories are the heaviest contributors to the $2.26M revenue, allowing Alfido Tech or the Superstore management to allocate future marketing budgets efficiently.

4.Profitability & Discount Impact: While generating $2.26M in top-line revenue is a strong indicator of market demand, evaluating the Net Profit alongside it is crucial. Analyzing how varying discount rates across different product lines impact overall profit margins will help the business balance competitive pricing with sustainable, long-term financial health.

5.Customer & Geographic Segmentation: Breaking down the 4,922 orders by customer segments (e.g., Consumer vs. Corporate) and geographic regions highlights where the highest-value demand originates. Identifying these regional and demographic hotspots empowers management to tailor localized marketing campaigns and optimize supply chain logistics for peak-performing areas.